Project Initialization and Planning Phase

Date	05 July 2026
Team ID	XXXXXX
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop a machine learning-based system to predict loan approval quickly and accurately using applicant information.
Scope	The system assists banks by automating loan eligibility prediction through a Flask web application and a Random Forest model.
Problem Statement	
Description	Manual loan approval is time-consuming and may result in inconsistent decisions. Banks need a faster and more reliable prediction system.
Impact	Improves loan processing speed, reduces human effort, and supports better decision-making for financial institutions.
Proposed Solution	
Approach	Use a Random Forest machine learning model integrated with a Flask web application to predict loan approval based on applicant details.

Key Features	Loan eligibility prediction Random Forest prediction model User-friendly web interface Fast decision support Real-time prediction results

SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/RAM	Intel Core i5
Memory	RAM	8 GB
Storage	Disk space for data, models, and logs	10 GB free space
Software		
Frameworks	Backend	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Joblib
Development Environment	IDE	VS Code / Jupyter Notebook
Data		

Data	Source, size, format	Kaggle dataset(Loan Prediction Dataset